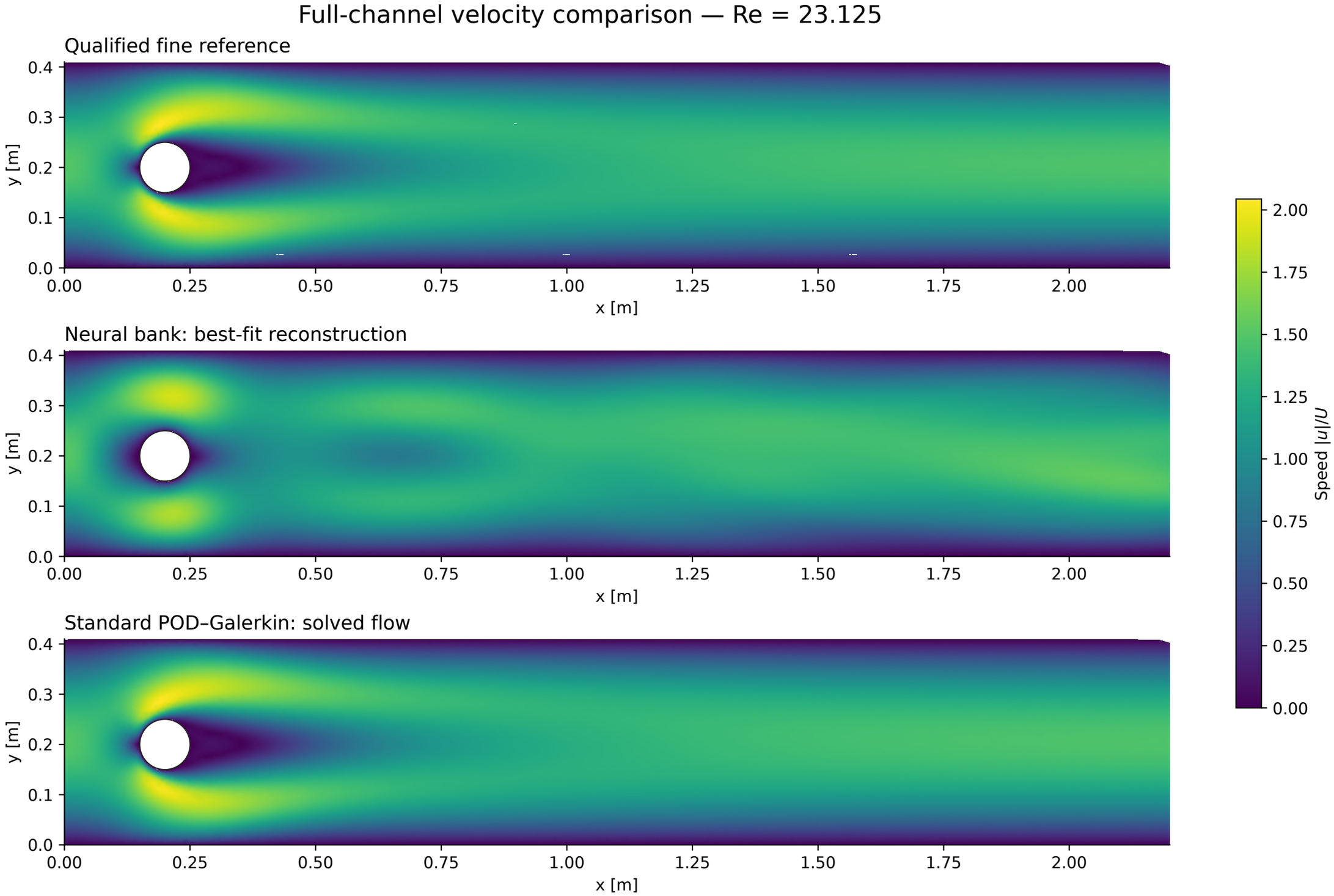

Full-channel velocity comparison — Re = 23.125
Qualified fine reference
Neural bank: best-fit reconstruction
Standard POD–Galerkin: solved flow
y [m]
x [m]
0.0
0.1
0.2
0.3
0.4
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Speed |u|/U
Same speed scale throughout. Neural result is best-fit projection; POD result is an actual Galerkin solve.